

★ WEEK 10 · PART 3 of 3

Regimes in Practice

How Many States, and Trading With Them

Choosing the number of regimes · Stage IV · the cautions

Fitting an HMM raises real choices: how many regimes, how to read them live, and how to act on them. We answer these, add regime-awareness to the pairs trade, and confront the traps of latent states.

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PART 3 · IN PRACTICE

Choosing and Using Regimes

How many states to fit, reading regimes live for trading, pairs trading Stage IV that stands down in crises, and the cautions that come with latent-state models.

From Algorithms to Decisions

Parts 1-2 built it; Part 3 uses it well

1

We can fit and decode HMMs

Forward for likelihood, Viterbi for the path, Baum-Welch to learn. The machinery is in place.

2

Now come the practical choices

How many regimes? How do we read them in real time without look-ahead? And how should a strategy act on them?

3

And the pairs thread gains a fourth stage

Stage IV: use the regime to decide WHEN to trade — standing down when the equilibrium is likely to break.

Part 3 in one line: choose the regimes wisely, trade them carefully, and respect the caveats.

How Many Regimes?

A modeling choice, not a fact of nature

The number of hidden states is not given by the data — you choose it. Too few and the model is crude; too many and it overfits, inventing regimes that are just noise. The count is a judgment call.

The trade-off

- More states fit the training data better, always.
- But extra states can capture noise, not structure.
- The goal is the fewest states that tell a real story.

How to decide

- Penalized likelihood (AIC/BIC) balances fit and complexity.
- Interpretability: can you name each regime?
- Out-of-sample stability of the fitted regimes.

Choosing the Number of States

Let a penalized likelihood guide you

$$\text{BIC} = -2 \log L + k \log n \quad (\text{penalize extra states; pick the lowest})$$

How BIC works

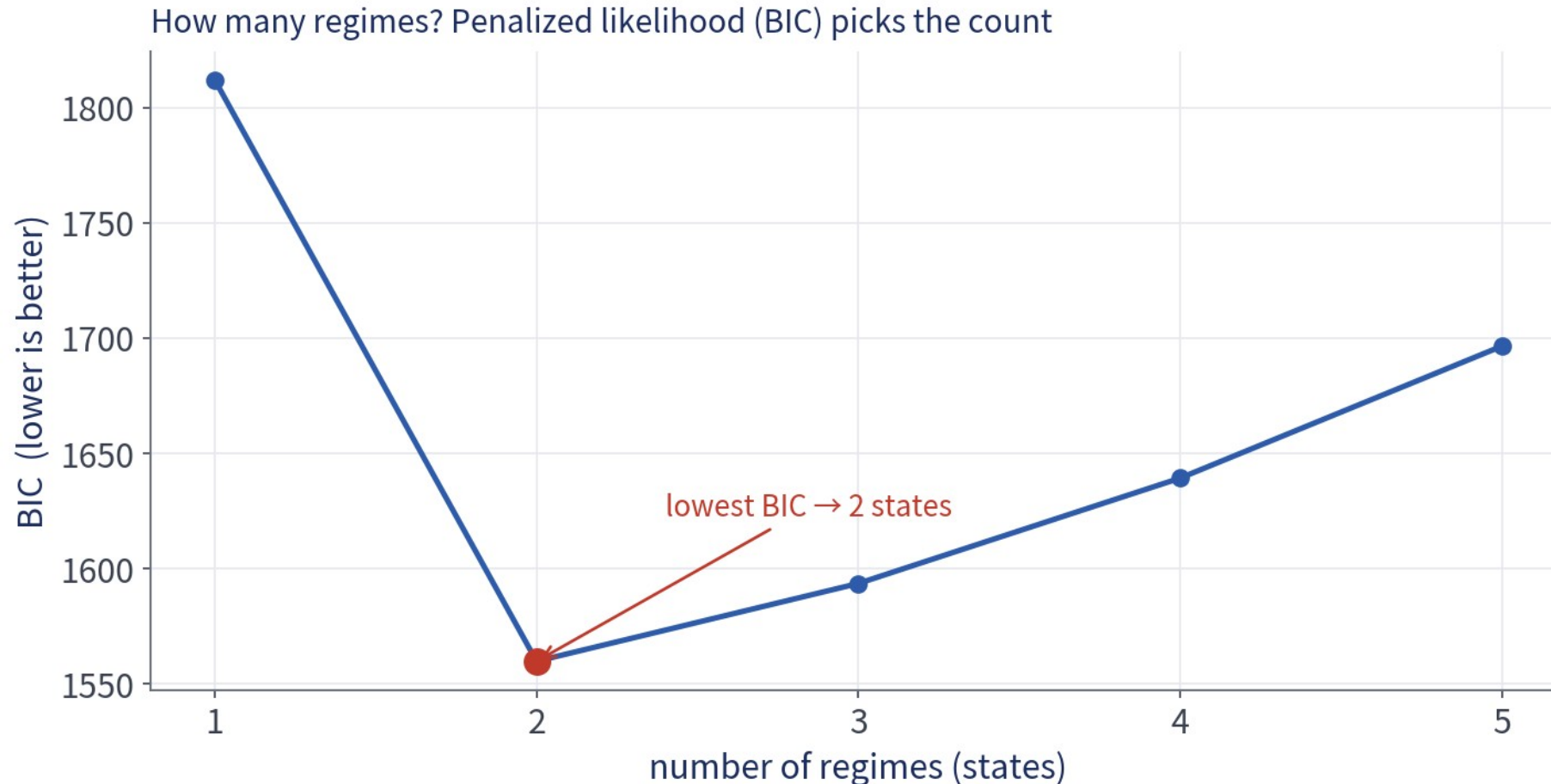
- It rewards fit (higher likelihood) ...
- ... but penalizes each added parameter.
- Pick the number of states with the lowest BIC.

Beyond the number

- Don't follow BIC blindly — check interpretability too.
- A statistically 'better' model with unnamed regimes is suspect.
- Simplicity and stability often win in practice.

How Many Regimes? Ask BIC

Fitting HMMs with one to five states and scoring each by BIC: the criterion drops then rises, and its lowest point suggests the right number of regimes for this data.



The Two-State Default

Calm and crisis, and why it's popular

Why two states

- Calm and crisis is the simplest useful split.
- It is easy to name, interpret, and act on.
- It captures the biggest feature: volatility regimes.

Its strengths

- Robust and hard to overfit.
- The regimes are almost always interpretable.
- A strong default before reaching for more.

When to Use More States

Three or more, and the risks

Sensible extensions

- Three states: bull, bear, and crisis, say.
- Or expansion, slowdown, and recession for macro.
- Use more only when each regime is meaningful.

The risk

- Extra states can splinter into noise regimes.
- Harder to interpret and less stable out of sample.
- Prefer the smallest count that stays interpretable.

Interpreting the Fitted Regimes

Label by behavior, not by index

How to label

- The states come out unlabeled, as 'state 0' and 'state 1'.
- Name them by their fitted mean and volatility.
- High-volatility, negative-mean → the crisis regime.

A caution

- The labels can swap between fits (label switching).
- Always identify regimes by their parameters, not their index.
- Check the transition matrix: which regime is sticky?

Regime Detection in Practice

Turning a fitted HMM into a live read

In production, you don't just decode history — you track the regime as data arrives. The filtered probability of each regime, updated every day, becomes a live gauge of the market's current mood.

The live signal

- Compute the filtered probability of each regime each day.
- Raise an alert when the crisis probability crosses a threshold.
- It updates causally, with no look-ahead.

Why filtered, not smoothed

- Smoothing uses future data — unavailable in real time.
- Live decisions must rely on the filtered estimate.
- Reserve smoothing for after-the-fact analysis.

Filtered vs Smoothed, for Trading

A subtle but crucial distinction

The trap

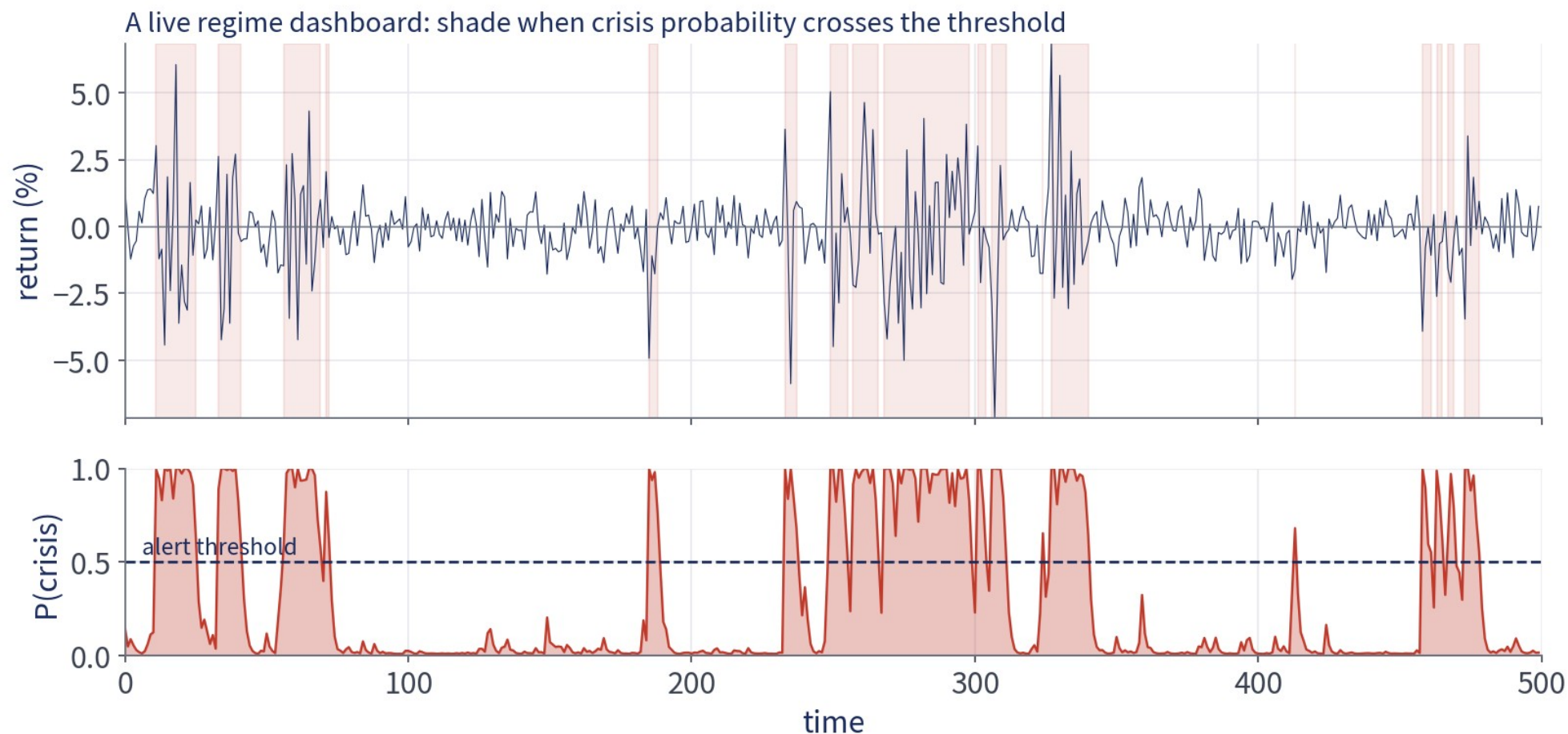
- Smoothed regimes look cleaner and more decisive.
- But they use the whole sample, including the future.
- Back-testing on smoothed regimes is look-ahead bias.

The rule

- Trade on FILTERED regime probabilities only.
- They are what you'd truly have known at the time.
- Use smoothing to explain, never to trade.

A Live Regime Dashboard

The filtered probability of crisis, updated each day. When it crosses the alert threshold, the periods are flagged (shaded) — a real-time read that never peeks at the future.



Stage IV: Regime-Aware Trading

The thread learns when NOT to trade

The pairs trade has bands (I), an anchor (II), and an adaptive hedge (III). Stage IV adds judgment: use the regime to decide whether to trade at all. In a crisis regime, the equilibrium may break — so stand down.

The addition

- Run a regime HMM alongside the spread.
- In the calm regime, trade the spread as usual.
- In the crisis regime, cut size or pause entirely.

Why it helps

- Cointegration tends to break exactly in crises (Week 7).
- Trading blindly then is how pairs strategies blow up.
- Knowing when to sit out protects the whole strategy.

Trade Only in Favorable Regimes

The regime as a position switch

Stage IV rule: $\text{position} = \text{base} \times P(\text{calm} \mid \text{data}) \Rightarrow \text{stand down in crisis}$

The simplest version

- Scale position size by the probability of calm.
- Full size in clear calm; nothing in clear crisis.
- A smooth dial rather than a hard on/off switch.

Variations

- A hard rule: trade only if crisis probability is low.
- Widen stops or shrink size as risk rises.
- The regime filter becomes a risk overlay.

Why Cointegration Breaks in Crises

The danger Stage IV guards against

What happens in a crisis

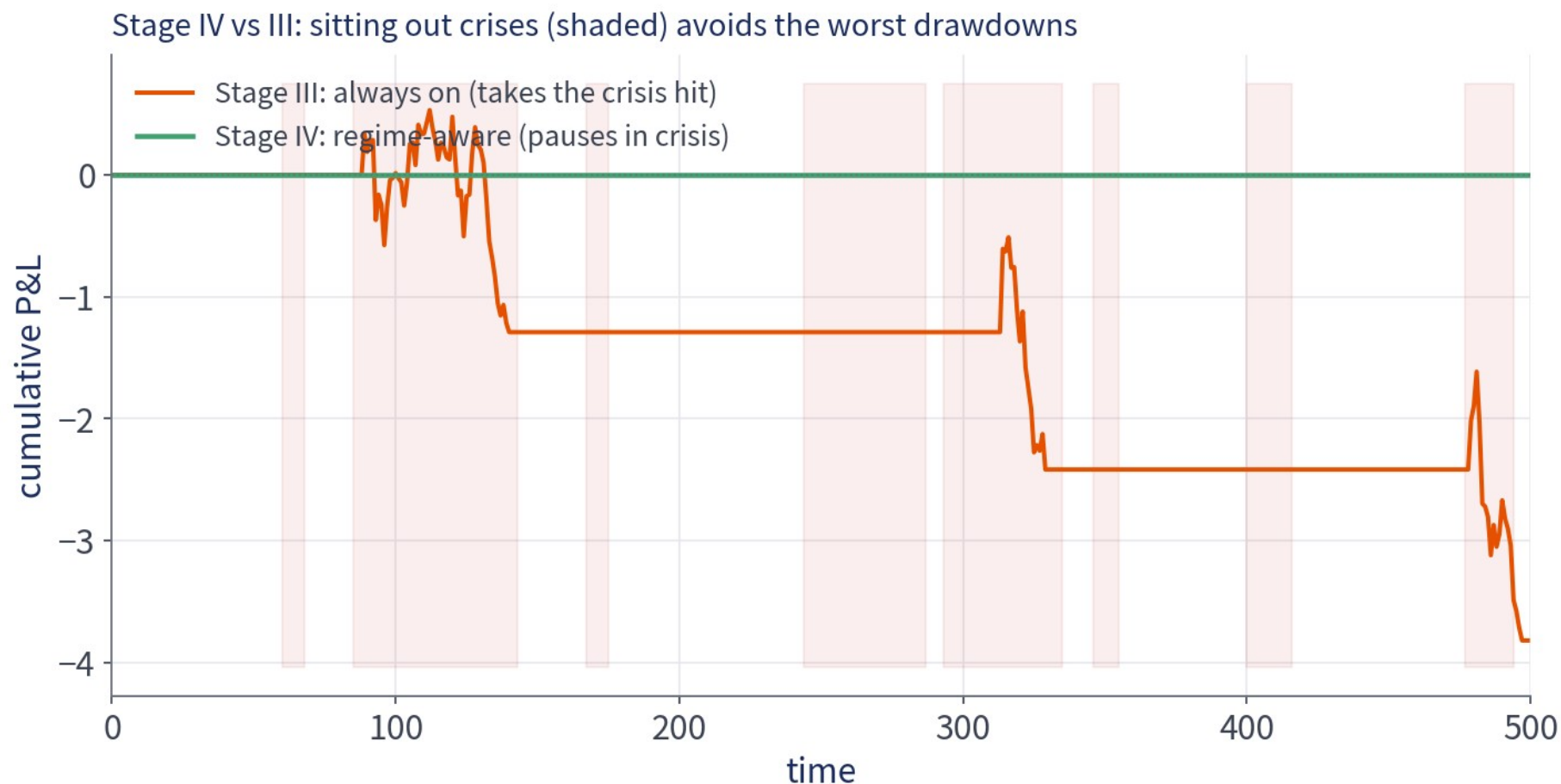
- Correlations and relationships shift abruptly.
- The historical equilibrium can simply stop holding.
- The spread drifts instead of reverting (Week 7).

How the regime filter helps

- It flags the crisis regime as it emerges.
- The strategy stands down before the breakdown compounds.
- It turns a known failure mode into a managed one.

Stage IV vs Stage III

An always-on strategy (orange) takes the full hit when the equilibrium breaks in a crisis (shaded). A regime-aware strategy (green) sits out those periods, avoiding the worst drawdowns.

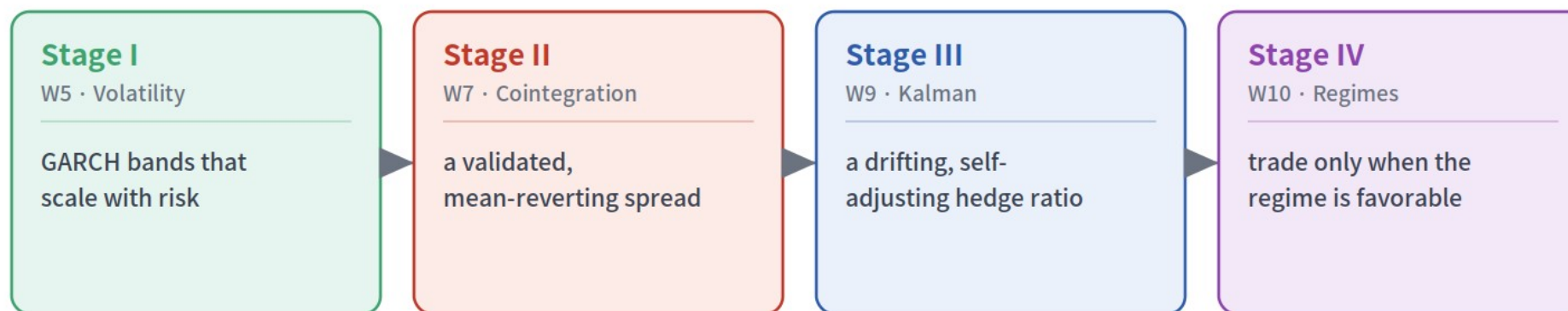


The Pairs Journey, Now Four Stages

Volatility bands (W5), a cointegration anchor (W7), a Kalman-adaptive hedge (W9), and now a regime filter (W10) that decides when to stand down. Knowing when not to trade completes the craft.

The Pairs Journey, Now Four Stages

Each week added a layer; Stage IV adds the wisdom to know when NOT to trade.



The lesson of Stage IV

Cointegration relationships tend to break precisely in crises (Week 7). A regime filter detects that danger and stands the trade down - or sizes it by the probability of calm - instead of trading blindly into a breakdown.

Knowing when NOT to trade is as valuable as the signal itself.

Cautions With HMMs

Latent states are seductive but slippery

An HMM will always find regimes — even in random data. The story it tells is compelling, which is exactly the danger. Latent-state models demand extra skepticism before you trust or trade them.

The seduction

- A fitted HMM produces a clean regime narrative.
- It looks insightful even when it is fitting noise.
- Compelling stories are the easiest to over-trust.

The discipline

- Validate regimes out of sample, not just in-sample.
- Check they are stable and economically sensible.
- Be ready to conclude the regimes aren't real.

Overfitting Regimes

More states, better story, worse model

The symptom

- Adding states keeps improving the in-sample fit.
- New regimes appear that have no real meaning.
- The model memorizes noise as 'structure'.

The guard

- Penalize complexity (BIC) and prize interpretability.
- Insist that regimes persist out of sample.
- When in doubt, use fewer states.

Label Switching and Identifiability

The regimes have no fixed names

The issue

- Which state is 'regime 0' is arbitrary and can flip between fits.
- So results keyed to the index are unstable.
- Two runs may 'disagree' only by relabeling.

The fix

- Identify regimes by their parameters (mean, volatility).
- Sort or pin states by, say, volatility before comparing.
- Never rely on the raw state index.

Regimes Are a Model, Not Reality

The map is not the territory

The reminder

- 'Calm' and 'crisis' are useful fictions, not physical states.
- The market doesn't truly toggle a two-position switch.
- The HMM is a convenient approximation of messier dynamics.

Why it matters

- Don't over-interpret a decoded regime as ground truth.
- Treat regime signals as one input, not gospel.
- Keep the model's limits in view when acting on it.

When an HMM Is the Wrong Tool

Match the model to the dynamics

Poor fits

- A truly continuous, smoothly-drifting state (use Kalman).
- Dynamics with long memory beyond one step.
- Cases where regimes can't be named or validated.

When it shines

- Discrete, persistent regimes like calm vs crisis.
- Volatility clustering with abrupt switches.
- A risk overlay that says when to stand down.

Regime-Aware Pairs Trade (Stage IV)

Direct Claude Code to trade only in calm regimes

GOAL Overlay a regime HMM on a pairs trade and back-test trading only when the calm regime is likely.

[INPUT] Prompt · Claude Code

Write and run `pairs_stage4.py` that:

1. fits a 2-state HMM to the spread's returns (or to a market index) to get FILTERED $P(\text{crisis})$;
2. back-tests the Stage III z-score pairs trade TWICE: always-on, and regime-aware (trade only when $P(\text{crisis})$ is below a threshold);
3. plots cumulative P&L for both, shading crises;
4. reports drawdown and Sharpe for each.

Use FILTERED (not smoothed) regimes to avoid look-ahead. Show the regime overlay cuts drawdown.

How to work with it

- Use filtered regimes only — no future data.
- Regime-aware should have smaller drawdowns.
- Compare Sharpe, not just total P&L.
- This completes the pairs thread (Stages I-IV).

Deliverable: `pairs_stage4.py` + always-on vs regime-aware P&L, with drawdown and Sharpe.

Three Questions Before You Finish

If you can answer these, Week 10 has landed

Q1	How do you choose the number of regimes?	BIC + interpretability
Q2	For live trading, filtered or smoothed regimes — and why?	Filtered (no look-ahead)
Q3	What does Stage IV add to the pairs trade?	Stand down in crises

All three come from model selection, the look-ahead rule, and regime-aware trading.

Three Pitfalls to Avoid

Where regime trading goes wrong

Common mistakes

- "More regimes means a better model."
- "Back-test on the smoothed regime path."
- "The decoded regime is the market's true state."

The reality

- Extra regimes often fit noise; prefer the fewest that work.
- Smoothed uses the future — trade on filtered only.
- Regimes are a useful model, not ground truth.

Week 10 in Three Ideas

Pulling the whole week together

1

A hidden regime, seen through returns

Parts 1-2: a Markov chain of latent regimes emits returns; the forward algorithm, Viterbi, and Baum-Welch evaluate, decode, and learn it — the discrete cousin of the Kalman filter.

2

Choose few regimes, read them live

Part 3: pick the number of states by penalized likelihood and interpretability, and track them with FILTERED probabilities to avoid look-ahead in trading.

3

Stage IV — know when not to trade

A regime filter stands the pairs trade down when the equilibrium is likely to break. But regimes are a model, not reality — trust them with discipline.

Next — Week 11: continuous-time models and stochastic differential equations.